

Predicting Major Hurricane Development from Historical Storm Data

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Introduction

Tropical hurricanes are among the most destructive natural hazards, threatening coastal populations worldwide. Hurricanes form over warm ocean water when heat and moisture cause air to rise and lower surface pressure. As air continues to rush toward the low-pressure center, the system begins to rotate and intensify. Hurricanes are classified from Category 1 through Category 5 based on maximum sustained wind speed, with Category 5 representing the most extreme storms.

Central pressure plays a critical role in this process because lower pressure is directly associated with stronger pressure gradients and higher wind speeds. As a result, wind speed and pressure together serve as fundamental indicators of hurricane intensity.

This project uses historical storm data to study how meteorological factors, such as central pressure and wind speed, and geographic characteristics, including storm location and timing, influence whether a storm intensifies into a major hurricane (Category 3–5). By building statistical and machine learning models, this study aims to improve understanding of hurricane behavior and support better forecasting, preparedness, and risk mitigation.

Objectives

The main objectives of this study are:

1. Predict major hurricane development (Cat 3–5) using early atmospheric indicators such as central pressure and maximum sustained wind speed.
2. Build models integrating environmental and geographic predictors to better understand hurricane behavior and classification.
3. Examine how interactions among key variables influence the prediction of major hurricane development.

Data and Materials

Data Sources

We use secondary data from historical hurricane (tropical cyclone) databases that include:

- Minimum central pressure
- Maximum sustained wind speed

- Storm category (e.g., tropical storm, Category 1–5)
- Track coordinates (latitude and longitude over time)
- Naming status (named vs. unnamed storms)
- Occurrence timing information (exact date, year, and season)

Data Preparation

Several data preparation steps were conducted prior to analysis:

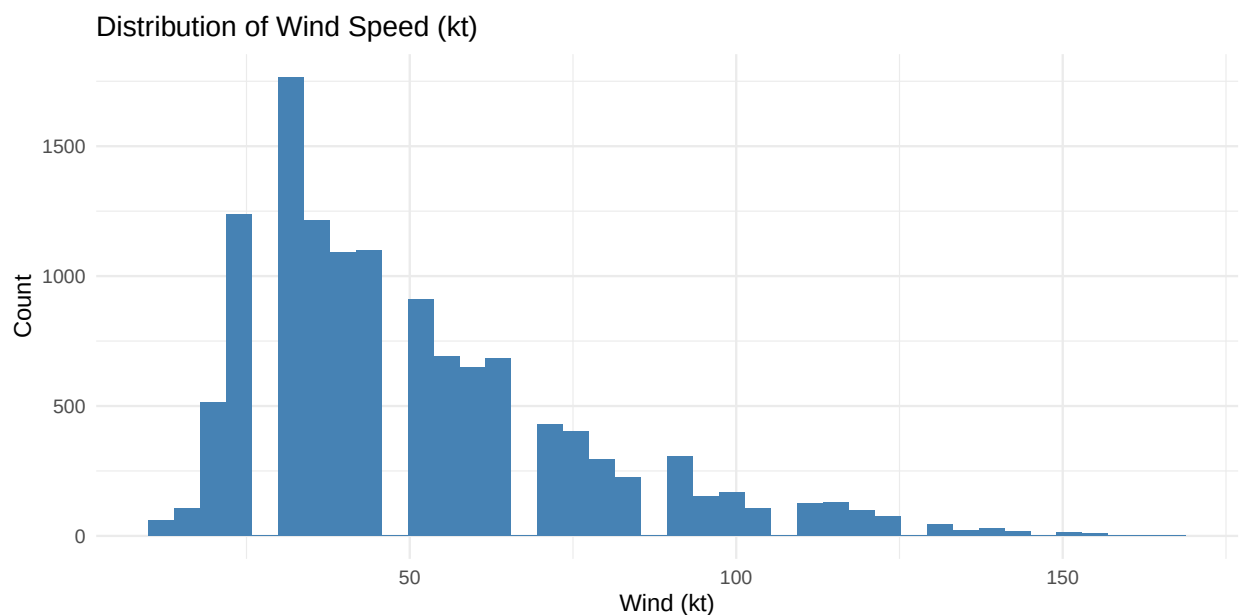
- Removal of missing, inconsistent, or physically impossible values. For example, storms recorded with zero wind speed or zero central pressure were excluded.
- Creation of derived variables, including a binary indicator for major hurricanes (major = 1 for Category 3–5, 0 otherwise).
- Construction of geographic region indicators using latitude and longitude.
- Extraction of timing variables and standardization of storm identifiers.

Additionally, although wind speed is a continuous physical quantity, the values in this dataset appear in discrete steps because operational hurricane wind estimates are reported in five knot increments. As a result, gaps in wind speed histograms reflect reporting conventions rather than true discontinuities in storm intensity.

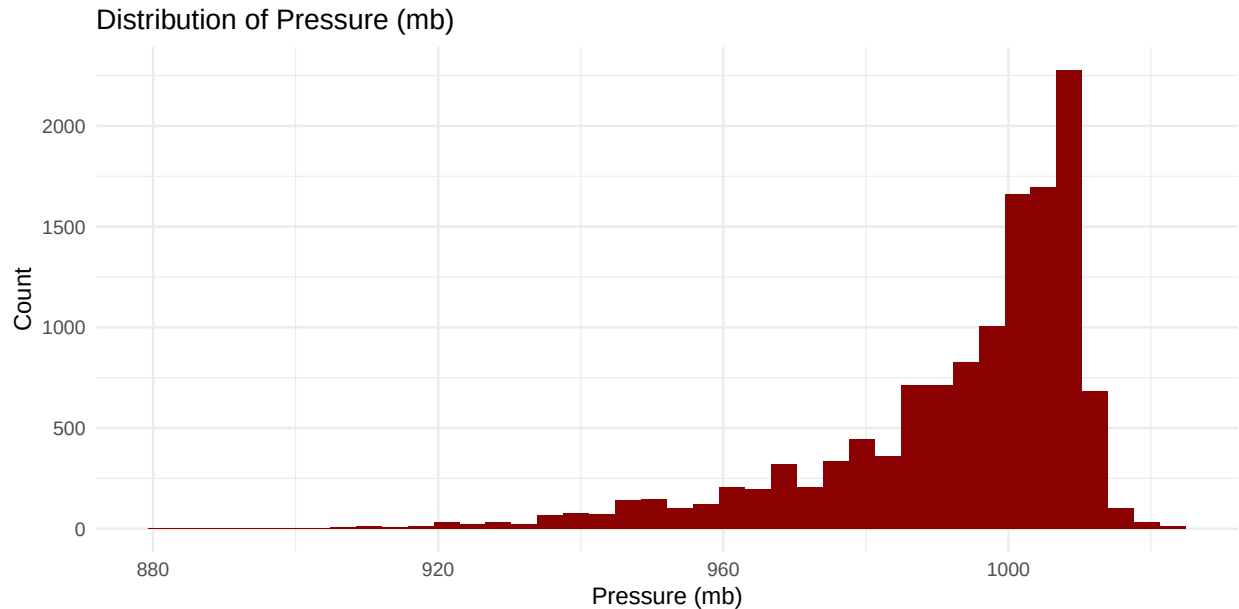
Exploratory Data Analysis

We begin with exploratory data analysis (EDA) to understand distributions of key variables and relationships between storm characteristics and major hurricane status.

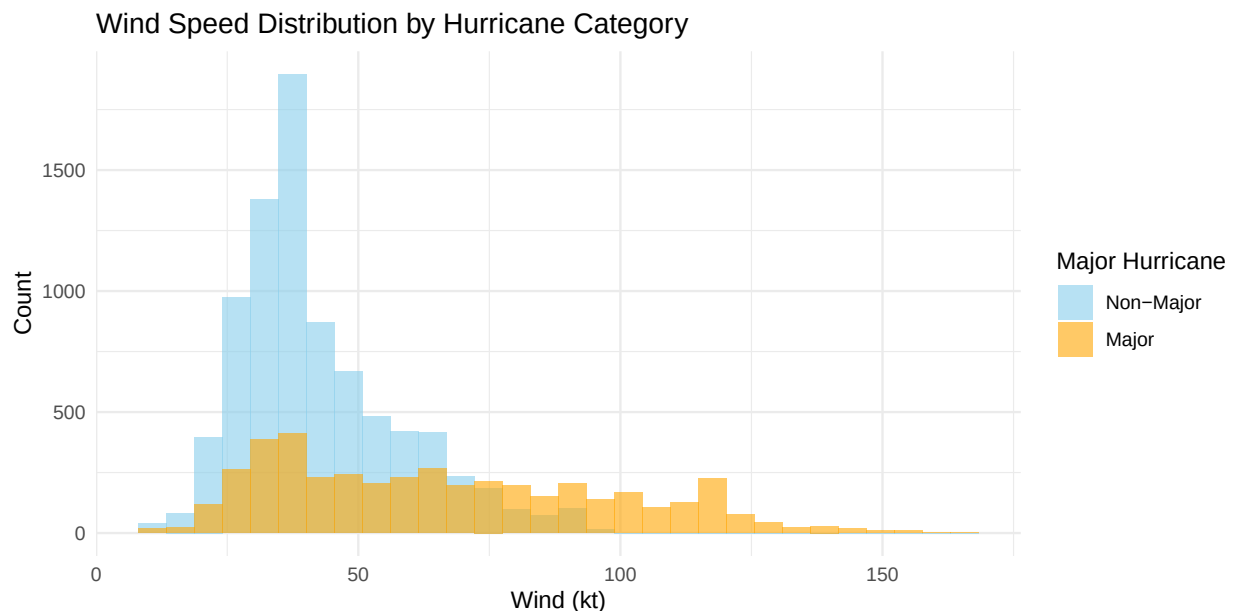
Histograms



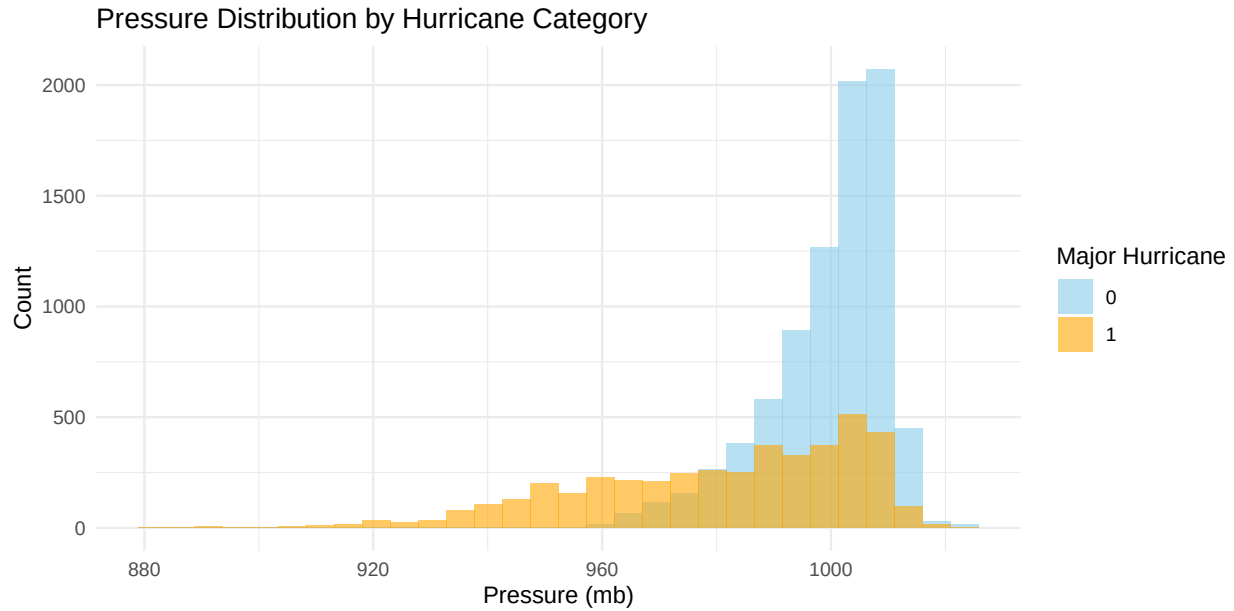
The wind speed histogram shows that most storm observations occur at lower wind speeds, with counts decreasing as wind speed increases. This is expected because most tropical storms do not intensify into strong hurricanes. The distribution is right skewed, with a long tail extending toward very high wind speeds. These higher wind speeds correspond to fewer but more intense storms, which are more likely to become major hurricanes.



The pressure histogram shows that most storms have relatively high central pressure, clustered near one thousand millibars. As pressure decreases, the number of observations drops. Very low pressure values are rare but correspond to the most intense storms. This pattern matches physical hurricane behavior, where stronger storms are associated with lower central pressure.

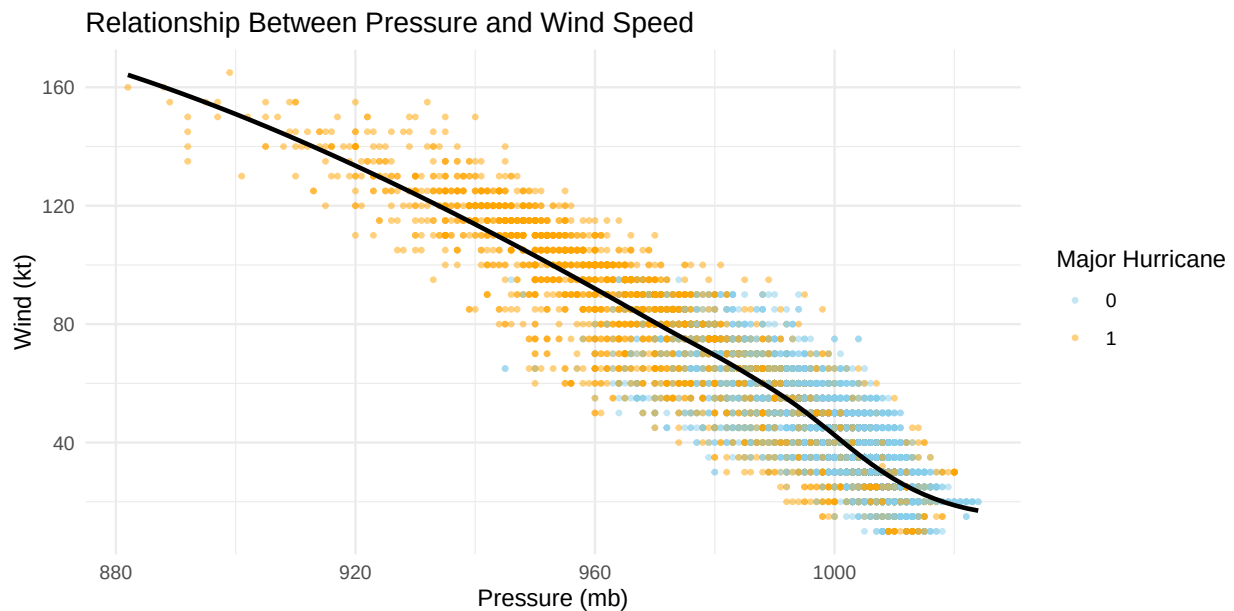


The wind speed distributions separated by major hurricane status further highlight these differences. Non major storms are mostly concentrated at lower wind speeds, while major hurricanes extend across much higher wind values. Although there is some overlap between the two groups, major hurricanes clearly dominate the upper range of wind speeds.



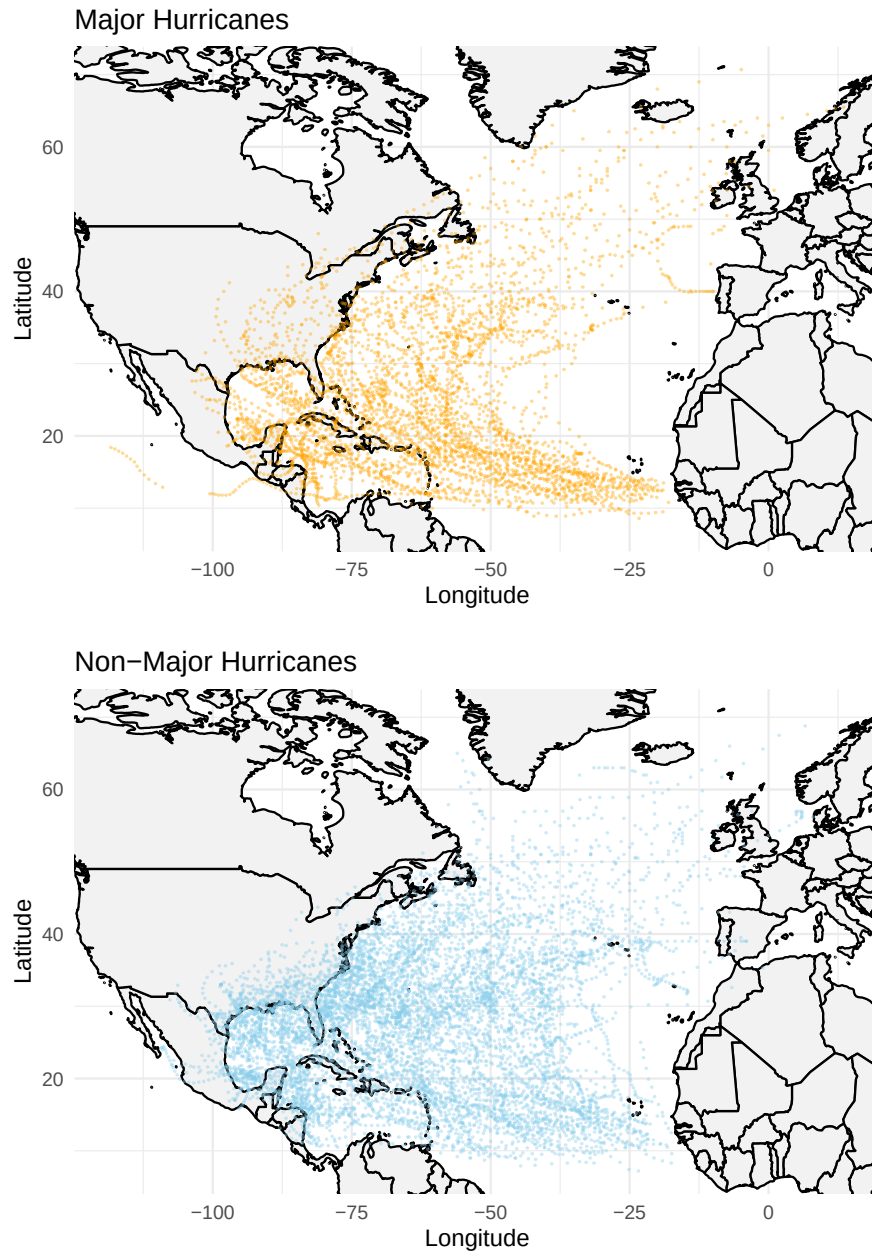
A similar pattern appears in the pressure distributions by hurricane status. Major hurricanes tend to occur at lower pressure values compared to non major storms. Non major storms are tightly clustered at higher pressures, while major hurricanes show a broader spread that extends downward toward much lower pressure levels.

Scatterplot



The scatterplot of pressure versus wind speed shows a clear negative relationship. As pressure decreases, wind speed increases. Major hurricanes are concentrated in the low pressure and high wind region of the plot, while non major storms mostly appear at higher pressure and lower wind values. This indicates that wind speed and pressure together provide strong signals of storm intensity.

Maps



The geographic maps provide additional context for storm behavior. Both major and non major hurricanes occur within similar latitude and longitude ranges, primarily over warm ocean regions. However, major hurricanes appear more concentrated in regions where environmental conditions are favorable for sustained intensification. These maps suggest that location alone does not determine intensity, but geography interacts with atmospheric conditions to influence hurricane development.

Overall, the exploratory analysis shows clear and consistent differences between major and non major hurricanes. Higher wind speeds and lower central pressure are strongly associated with major hurricane formation. The joint relationship between wind and pressure, along with geographic patterns, supports the use of interaction terms and nonlinear models in later analysis.

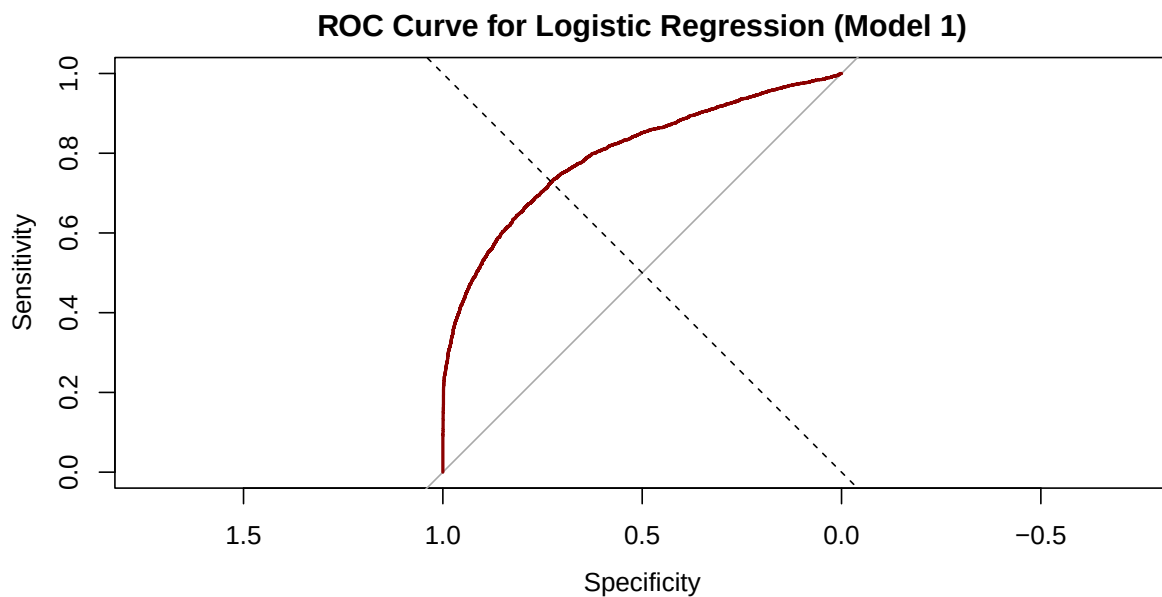
Methods

This study uses supervised classification models to predict whether a storm becomes a major hurricane.

Logistic regression models were first used as a baseline parametric approach. An initial model included wind speed, pressure, latitude, longitude, and region as main effects. A second logistic model incorporated interaction terms between wind speed and pressure and between wind speed and latitude to better capture meteorological dynamics. Model performance was evaluated using ROC curves, AUC, confusion matrices, and sensitivity–specificity tradeoffs. Classification thresholds were optimized using Youden’s J statistic.

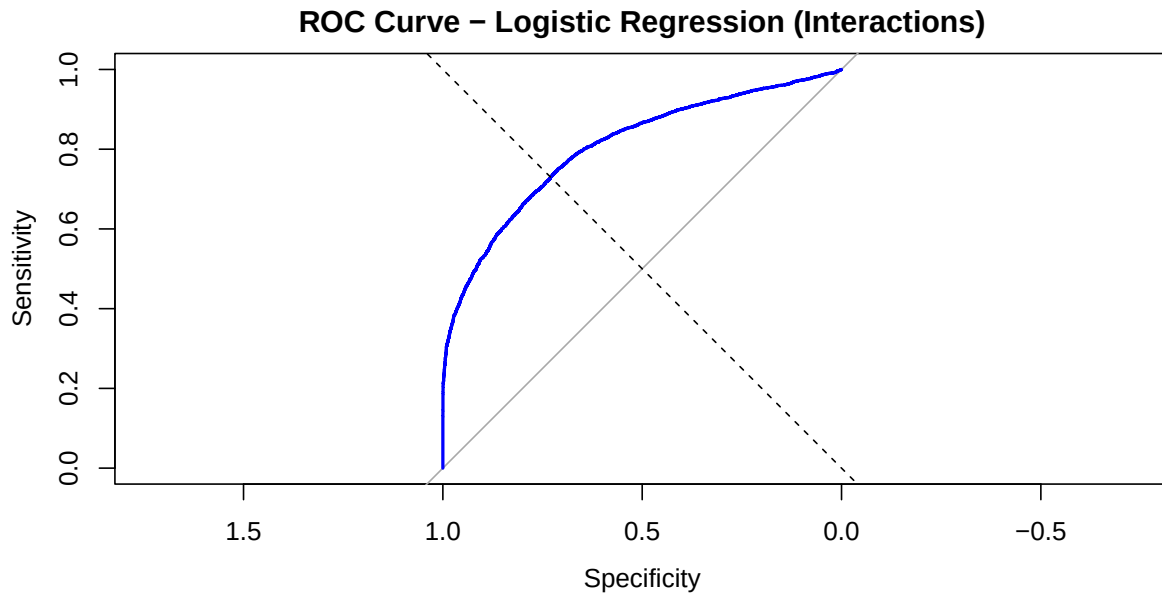
Model 1: Logistic Regression (Main Effects Only)

Area under the curve: 0.7975



Model 2: Interaction effects

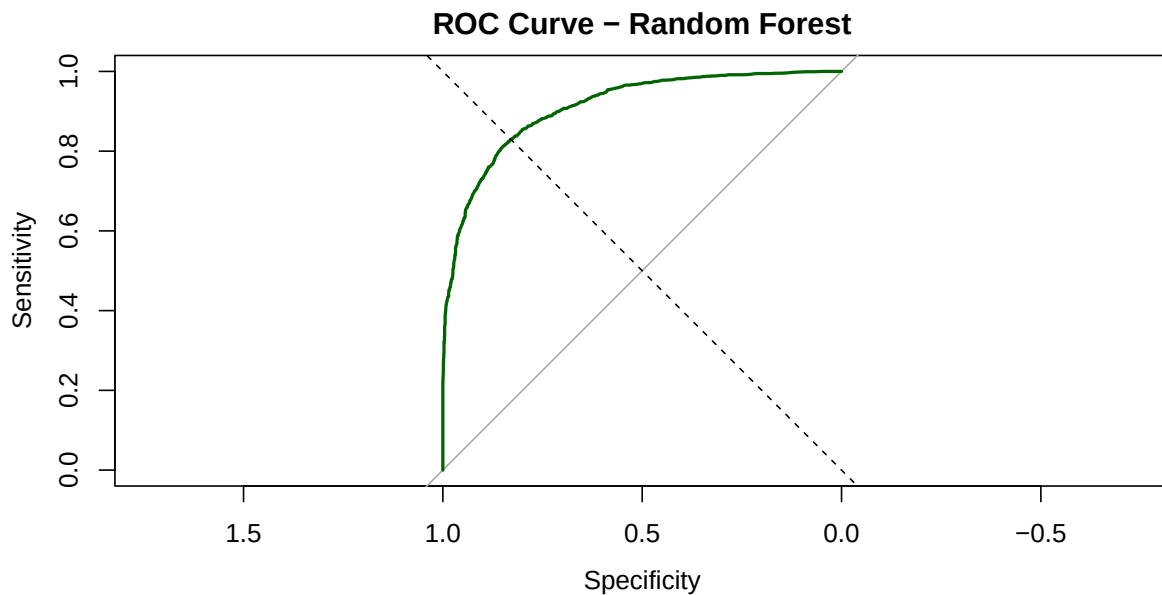
Area under the curve: 0.8052



Random Forest models were then employed to capture nonlinear relationships and complex interactions automatically. A baseline Random Forest using core predictors was compared to a model that included manually engineered interaction features. Performance was assessed using a train-test split, ROC curves, AUC values, confusion matrices, and variable importance measures.

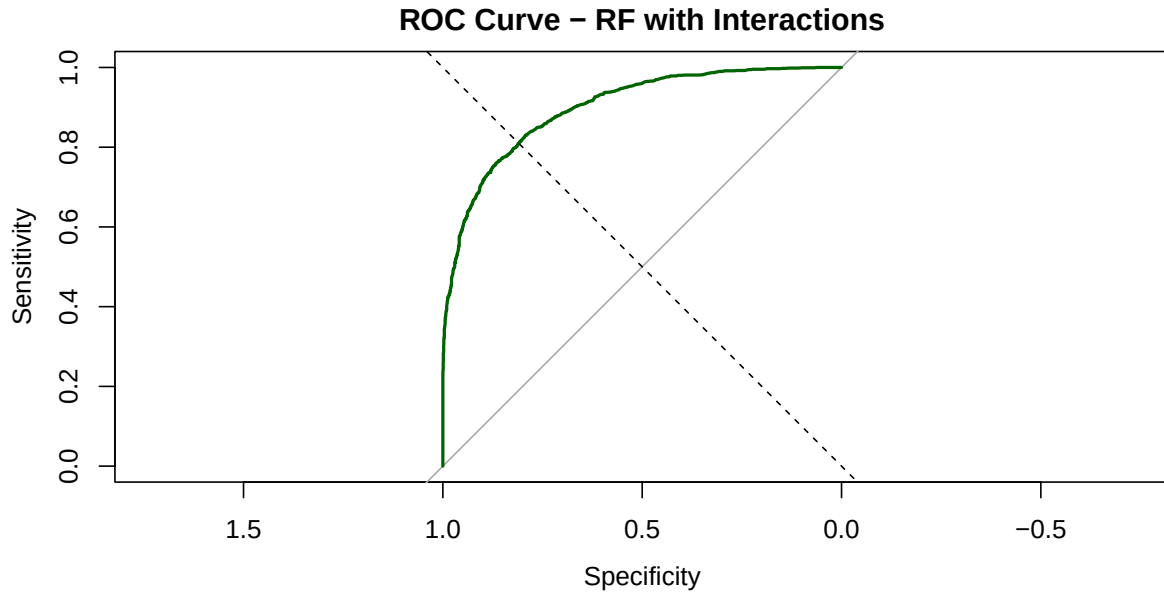
Model 3: Random Forest (Main Effects Only)

Area under the curve: 0.9109

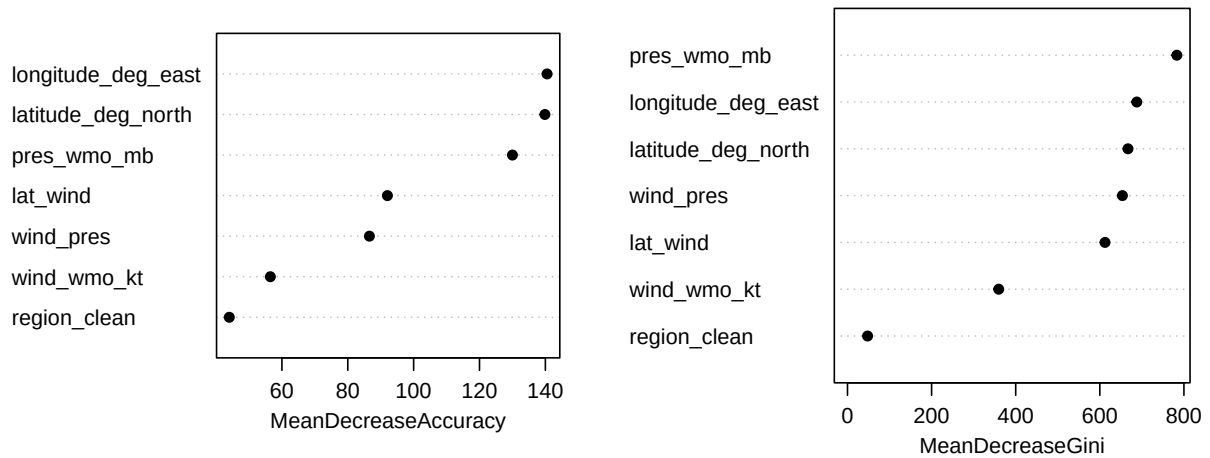


Model 4: Random Forest with Interaction Features (wind×pressure, latitude×wind)

Area under the curve: 0.8994



Variable Importance – Random Forest with Interaction Effects



Results and Interpretation

Model 1 used a logistic regression with only main meteorological and geographic effects. It achieved an AUC of 0.7975, indicating moderate discrimination. At the default 0.5 threshold, the model showed high specificity but low sensitivity, correctly identifying most non major storms while missing more than half of major hurricanes. This suggests that linear main effects alone are insufficient to capture the complexity of hurricane intensification.

Model 2 extended the logistic regression by adding interaction terms between wind speed and pressure and between wind speed and latitude. Model performance improved modestly, with AUC increasing to 0.8052. Using the optimal threshold increased sensitivity from 0.48 to 0.67, while specificity declined to 0.79, resulting in a higher balanced accuracy. This trade off reflects improved detection of major hurricanes at the cost of some false positives. The results indicate that interactions among meteorological variables play an important

role in hurricane intensification.

The Random Forest model using only core predictors substantially outperformed both logistic regression models. It achieved an AUC of 0.9109 with balanced sensitivity and specificity, correctly identifying over 80 percent of major hurricanes. Variable importance measures showed that wind speed, pressure, and latitude were the most influential predictors, highlighting the joint and nonlinear nature of hurricane development.

Adding manually engineered interaction features to the Random Forest did not improve performance and slightly reduced AUC to 0.8994. This outcome is expected because Random Forests naturally capture interactions through their tree structure. Overall, the baseline Random Forest provided the best predictive performance, demonstrating that hurricane intensification is governed by nonlinear and interacting meteorological factors that are better modeled using flexible machine learning approaches.

Limitations

This study has several limitations related to the data and the models used. Some important weather factors, such as sea surface temperature and humidity, were not available, which limits how well storm behavior can be explained. Additionally, older storm observations may be less accurate because measurement technology was less advanced in the past. Major hurricanes also occur less frequently than non major storms, creating an imbalance in the data that can affect model performance. Data cleaning steps, such as removing missing or physically impossible values, may exclude unusual storms and reduce the overall diversity of the dataset. From a modeling perspective, logistic regression has difficulty capturing complex relationships, while Random Forest models, although more flexible, are harder to interpret. Lastly, because the models are trained on historical data, they may not fully reflect changes in hurricane behavior driven by evolving climate conditions.

Conclusion

This study demonstrates that hurricane intensification is driven by complex and nonlinear interactions among meteorological and geographic factors. Although logistic regression provides a useful baseline, its performance remains limited even when interaction terms are included. In contrast, Random Forest models capture these nonlinear relationships more effectively and achieve substantially higher predictive accuracy.

The strongest performance was achieved by the baseline Random Forest model using only core predictors such as wind speed, central pressure, and location. Its success without manually engineered interaction terms highlights the model’s ability to learn complex relationships directly from the data. These results indicate that flexible machine learning approaches are better suited for predicting major hurricane development than linear models.

From a practical standpoint, improved identification of storms likely to intensify can support earlier warnings and better preparedness efforts. These findings highlight the value of nonlinear modeling techniques for improving the understanding and prediction of hurricane behavior.

References

- Emanuel, K. (2021). Tropical cyclones and climate change. *Annual Review of Earth and Planetary Sciences*, 49, 1–32.
- Holland, G., & Bruyère, C. (2014). Recent intense hurricane trends in the context of climate change. *Nature*, 509, 349–352.
- Knutson, T., et al. (2019). Tropical cyclones and climate change assessment: Part I: Detection and attribution. *Bulletin of the American Meteorological Society*, 100(10), 1987–2007.